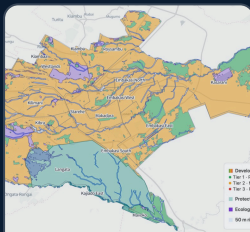


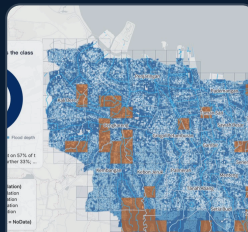
MapX built-in scenarios · urban and rural planning

Scenario Handbook

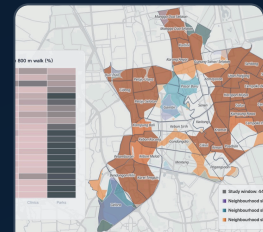
Six planning scenarios: study design, data and methods



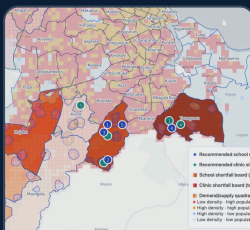
1. Development Suitability Baseline



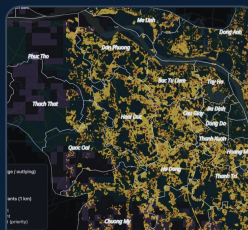
2. Carrying Capacity Assessment



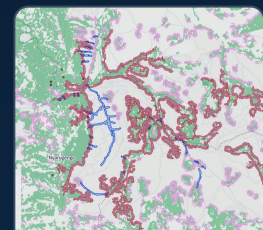
3. 15-minute Life Circle Coverage



4. Public Facility Siting



5. Urban Growth Monitor



6. Land-use Conflict Screening

About this handbook

This handbook turns the six built-in urban and rural planning scenarios in MapX into self-contained chapters: study design, data, methods and platform capabilities, how to run the scenario, and what one example run produced — with the limits stated.

Edition	First edition · October 2026
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Scope	The six built-in urban and rural planning scenarios in MapX
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Content	Scenario pages from the MapX documentation and the built-in data catalog
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Data	Each scenario's entry layers, with source, licence and attribution per layer
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Related docs	Scenarios, built-in datasets, analysis, plan mode, charts, reports
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Support	support@mapxagent.com
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Content and screenshots come from the MapX documentation and the built-in scenarios; map layer credits and licences are attached to each entry dataset. Example figures illustrate the shape of the deliverables only.

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Foreword

A scenario bundles everything a piece of spatial analysis needs before you type anything: a study area, the prepared datasets for it, a method that maps onto the platform's analysis tools, and suggested questions that work with that data. Opening one turns an empty project into a project that is ready to answer a real planning question.

What this handbook covers

This handbook collects the six built-in urban and rural planning scenarios in MapX. Each scenario has one chapter, and every chapter follows the same structure: the planning question, the data it ships, how the study works, the platform capabilities it uses, how to run it, what one example run produced, and the limits to state when you read the results.

Scenario	Study area	Layers	What it answers
Development Suitability Baseline	Nairobi	8	Which land is buildable once ecology, slope and flood constraints are applied
Carrying Capacity Assessment	Jakarta	6	How much development the water, climate and flood-hazard constraints allow
15-minute Life Circle Coverage	Jakarta	7	Who is covered by schools, clinics and green space within walking distance, and who is not
Public Facility Siting	Dar es Salaam	7	Where coverage gaps are, and which candidate sites close them
Urban Growth Monitor	Hanoi	7	How fast, in which direction, and how compactly the built-up area is expanding
Land-use Conflict Screening	Kigali	8	Where development collides with ecological space, steep slopes and riverbanks

How to read it

- For "where can the city build, and how much can it hold": start with chapters 1 (Nairobi) and 2 (Jakarta); chapter 6 (Kigali) applies the same constraint logic to built-up land.
- For public service coverage and facility siting: read chapter 3 (life circle) before chapter 4 (siting) — the first finds the gaps, the second proposes sites.
- For urban expansion and growth management: go straight to chapter 5 (Hanoi).
- Every chapter stands on its own; the Related list at the end of each chapter names the documentation topics to continue with.

■ About the numbers

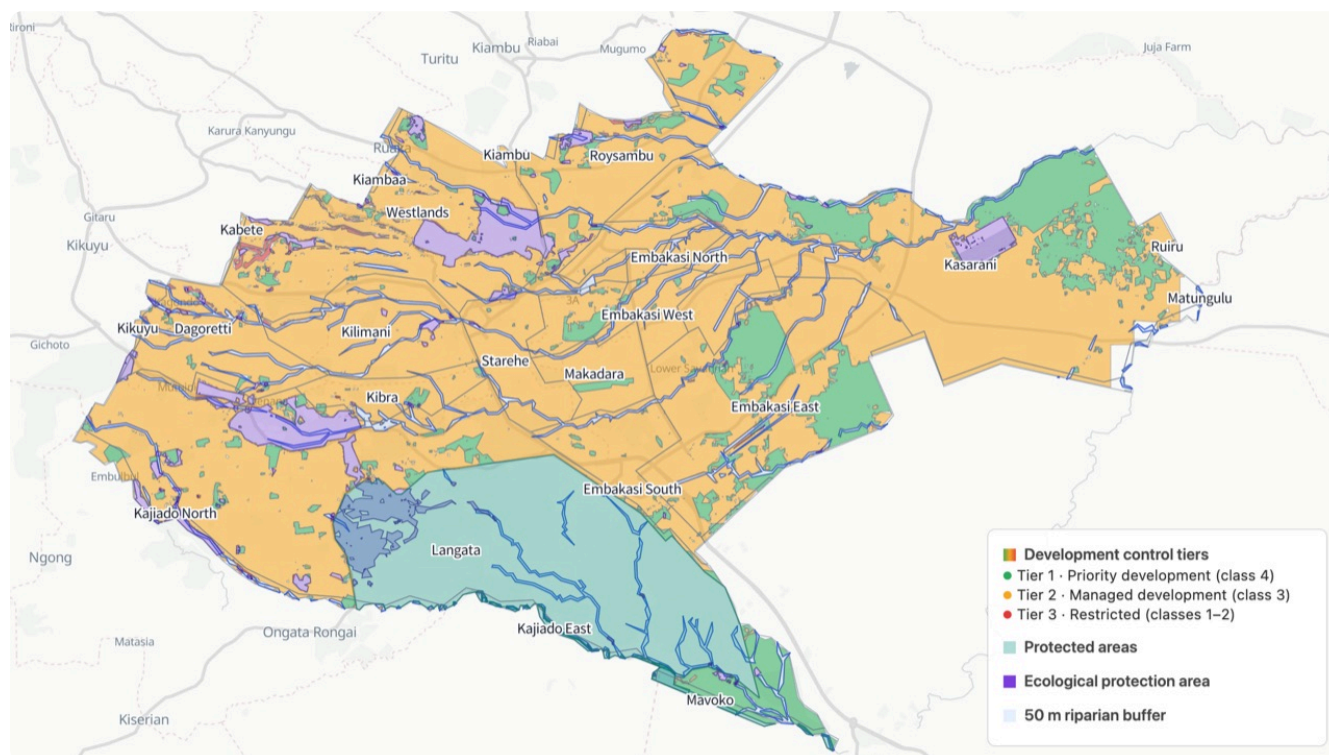
The areas, shares, rankings and site counts in this handbook come from one example run with the default settings on the shipped data. They show the shape and plausible magnitude of the deliverables, not what your run will return. Your study window, parameter choices and data version will change the values; each chapter's Example output box repeats this caveat.

■ Data and licences

Every entry layer in the six scenarios comes from the MapX built-in data catalog, with source, licence and attribution attached per layer. Scenarios are analysis aids: they deliver layers, charts and reports, and they do not replace statutory planning documents.

Development Suitability Baseline Nairobi

Screen ecological, terrain and flood constraints, grade the remaining land, estimate carrying capacity and propose a development boundary with tiered controls.



The Nairobi scenario as delivered: development control tiers drawn over ecological protection areas, protected areas and the 50 m riparian buffer

AT A GLANCE	
STUDY AREA Nairobi	ENTRY LAYERS 8 layers
KEY METHODS Hard-constraint screening · weighted overlay · capacity estimate	DELIVERABLES Suitability classes, constraint layers, capacity tables, charts and a report

Development Suitability Baseline answers the first question of any growth plan: **where can this city build, and how much can it hold** once ecological, terrain and flood constraints are respected. It follows the two-part appraisal logic used in Chinese territorial planning — hard constraints first, suitability and carrying capacity second, boundary and tiered controls last.

What the scenario ships

Opening the scenario imports eight prepared layers for the Nairobi window. Every layer comes from the **built-in datasets** catalog with its default style, legend, source, licence and attribution attached.

Layer	Data	Role in the study
Road network	52,642 OSM lines	Existing urban skeleton; reference for the boundary and for infill versus new growth
Watercourses	718 OSM lines	Source of the 50 m riparian buffer
Administrative boundaries	26 ADM3 units (geoBoundaries)	Attribution, summary tables and map labels
Protected area constraints	379 Overture features across Kenya	Part of the ecological exclusion
DEM	Copernicus DEM GLO-30, 30 m (1,459–1,954 m)	Slope classes
Land cover	Esri Sentinel-2 2020, 10 m, 9 classes (57% built-up)	Ecological classes and the existing built-up surface
Flood hazard	WRI Aqueduct, 1 km	Flood avoidance line
Population	GHS-POP, 1 km	Current population reference for the capacity comparison (hidden by default)

The layer order is the display order: points, then lines, then polygons, then rasters. All eight stay in the project for follow-up questions; only the surfaces you see in the legend above are styled as deliverables.

How the study works

The method is a chain of three planning questions. Each one maps onto deterministic **analysis** tools, so every number in the report can be traced back to a run.

Stage	Question	What MapX runs	Output
1. Screen the hard constraints	Which land is off-limits?	Slope, Reclassify, Buffer, Raster calculator, Overlay erase, Raster class area, Zonal statistics	Ecological protection mask and its polygons, the 50 m riparian buffer, constraint-area tables
2. Grade and size what remains	How suitable is the remaining land, and how many people can it carry?	Weighted overlay (equal weights by default), Reclassify, Grid at 1 km, Zonal statistics	Suitability classes 1–4, population per evaluation cell, class-area and capacity tables
3. Propose the boundary	Where should the development boundary run, and what tiers apply?	Raster to polygons, Overlay erase and dissolve, Zonal summary, Label anchors	Development control tiers (priority / managed / restricted), charts and a written report

The constraint definitions are explicit. Ecological protection is the union of protected areas, the water, tree and flooded-vegetation land-cover classes, and the 50 m watercourse buffer; slope is split at 8°, 15° and 25°; flood is an avoidance line, not a suitability score. Suitability is then graded 1–4 over the land that survives those exclusions.

The capacity estimate is a planning convention. Carrying capacity is derived at 100 m² of urban construction land per person. The report states the assumption; it is an estimate, not a measured capacity.

Platform capabilities

- **Plan mode** turns the request into a reviewable plan: the AI asks about the study area, which constraints are hard, and the report audience before anything runs.
- **Analysis tools** do the computation: slope and reclassification, raster calculator masks, buffering, weighted overlay, 1 km grids, zonal statistics and polygon conversion.
- **Layer styling** gives the control tiers human category labels; the constraint layers stay translucent so the tiers remain readable.
- **Charts** and a Word **report** package the class areas, the constraint breakdown and the capacity ranking for delivery.

Run it

1

Open the scenario

Go to **Scenarios** in the top menu, select **Development Suitability Baseline** and choose **Open scenario**. Eight layers land in the project and the camera flies to Nairobi (36.82, -1.29).

2

Send the suggested question

The scenario ships one starter prompt: *"Run a development-suitability study for Nairobi: screen the ecological, terrain and flood constraints, grade buildable land and carrying capacity, then propose a development boundary with tiered controls; deliver layers, charts and a report."* Edit it if your study area or constraints differ.

3

Answer the plan clarifications

Plan mode asks three things: the study area and admin unit, which constraints count as hard for this city, whether the ecological mask is a hard exclusion (the default) or a graded factor, and the report language. Everything else — grid size, slope thresholds, class breaks — is stated as an assumption.

4

Review and approve the plan

Read the plan document, adjust any step, then approve it. Each step runs as a deterministic tool with its own status, time and result; the run graph shows how the layers feed each other.

5

Refine and deliver

Turn on the population layer to compare present-day population with the estimated capacity, restyle the tiers, then export the report or share the map.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped Nairobi data, the deliverables were a map with three development control tiers, the ecological protection polygons, protected areas and the riparian buffer, plus:

- four suitability classes of **0.4 / 4.2 / 446 / 83.6 km²** after hard exclusions;
- a constraint breakdown of **105.5 km²** protected areas, **40.4 km²** riparian buffer, **39.1 km²** ecological composite, **5 km²** flood avoidance area and **0.04 km²** steep slope ($\geq 25^\circ$);
- a capacity ranking by sub-county — Kasarani, Langata and Embakasi East lead the estimate;
- three charts (buildable land by class, constraints screened, capacity by sub-county) and one report.

Read the results with care

▲ **Flood is an avoidance constraint here.** The shipped 1 km flood raster has only a few valid cells in this window; treat it as a "keep clear of the floodplain" line. A graded flood factor needs a different dataset.

▲ **Capacity is an estimate at 100 m² per person.** Report it as a planning assumption, never as measured capacity.

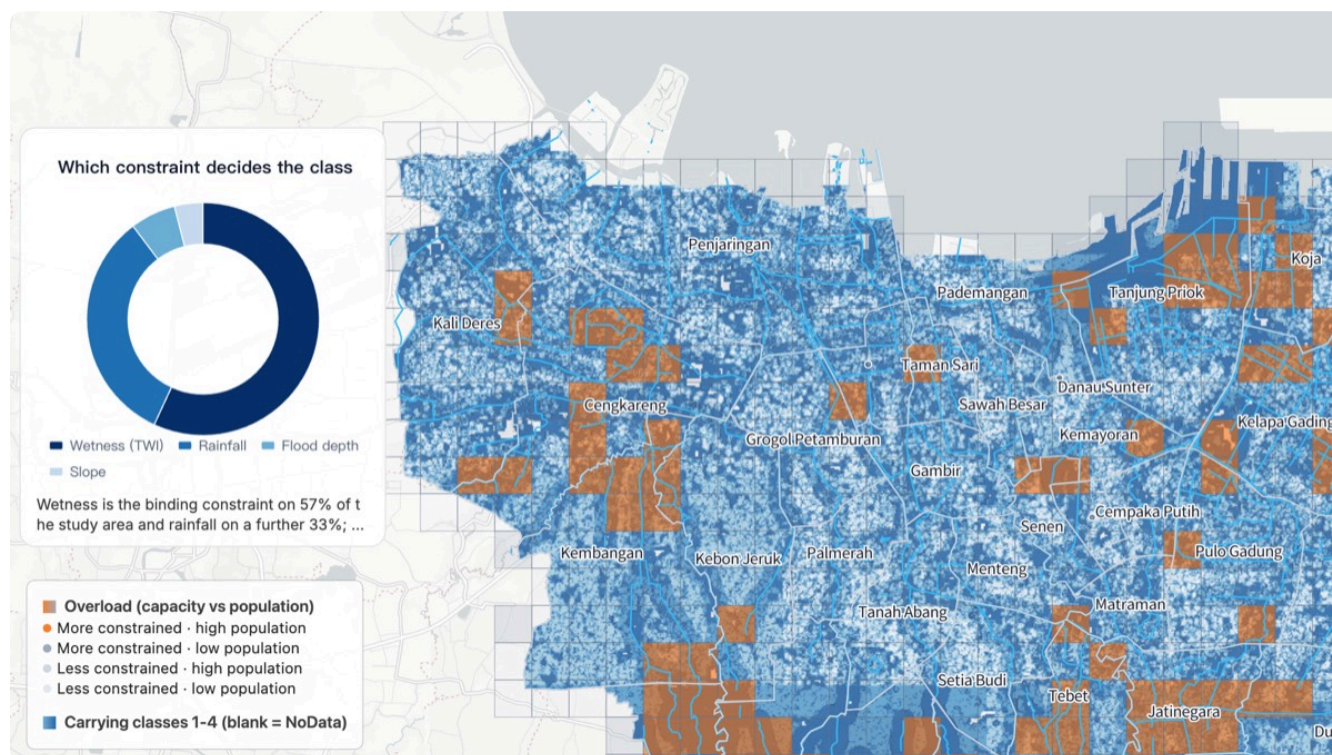
▲ **Population never enters the suitability score.** It is a present-day reference for comparing what exists with what the land could hold.

▲ **The ecological mask is a product of its inputs.** It uses protected areas plus three land-cover classes and a 50 m buffer; grass and shrub are deliberately excluded, and local statutory boundaries should replace the country-wide preview when you have them.

▲ **Equal weights are the default composite.** If you want AHP or expert weights, run the weighted overlay as a documented sensitivity variant and compare the class tables.

Carrying Capacity Assessment Jakarta

Assess how terrain, wetness, flood depth and rainfall constrain Jakarta's growth, grade carrying capacity, and flag the districts where pressure is highest.



The Jakarta scenario as delivered: carrying-capacity classes 1–4, overload cells and the constraint-composition chart

AT A GLANCE	
STUDY AREA Jakarta	ENTRY LAYERS 6 layers
KEY METHODS Fuzzy membership · limiting-factor composite · pressure quadrants	DELIVERABLES Capacity classes, overload cells, constraint tables, charts and a report

Carrying Capacity Assessment asks how much development a city can absorb before its water, climate and hazard constraints become the binding limit. For Jakarta — a low, flat delta city with frequent waterlogging — the study reads four constraints together, grades the result, and separates "constrained land" from "land under pressure".

What the scenario ships

Six prepared layers cover the Jakarta window. The rasters except rainfall start hidden so the class raster stays readable; turn them on when you want to inspect a single factor.

Layer	Data	Role in the study
River network	OSM waterways	Drainage and flood corridor
Administrative boundaries	46 kecamatan (OCHA COD-AB ADM3); 44 on the mainland	Study-area definition, attribution and district labels
Precipitation	CHIRPS, ~5.35 km	Rainfall membership (waterlogging constraint)
Population	WorldPop, 100 m (hidden)	Exposure and pressure axis
Flood hazard	WRI Aqueduct, ~900 m (hidden)	Flood-depth membership
Terrain	SRTM, 30 m (hidden)	Slope and topographic wetness

All six come from the **built-in datasets** catalog, with source, licence and attribution attached.

How the study works

The design follows the "shortest stave" reading of carrying capacity: the most limiting factor decides the class, not the average.

Stage	Question	What MapX runs	Output
1. Build the capacity classes	Which factor limits each place, and how strongly?	Slope, Topographic wetness index, Fuzzy membership, Raster calculator (<code>minimum</code>), Focal statistics repair, Euclidean distance, Reclassify	Four-class carrying-capacity raster with the 300 m drainage-corridor adjustment
2. Diagnose pressure	Where is low capacity combined with high population?	Grid at 1 km, Zonal statistics, Quadrant (median split)	Overload cells, population per cell, district pressure table
3. Explain and mitigate	Which constraint dominates, and what should each district do?	Raster class area, Zonal statistics, Charts, Report	Class-area and dominant-constraint tables, mounted pressure chart and report

Four memberships feed the composite. Slope is scored from 0–25°, wetness from the TWI range calibrated inside the study window (6.08–9.28), flood depth from 0–3 m and rainfall from 1770–2521 mm inside the same window. Rainfall is a *waterlogging* constraint: more rain means a stronger limitation, not a bigger resource.

Gaps are repaired, not hidden. The flood raster stops at its model domain and the wetness index is undefined on perfectly flat cells. Both are filled from their own neighbourhood mean and reported as a data repair; a coverage floor of 99% of the study area is the acceptance check.

Population is the pressure axis. It enters the overload quadrant and the capacity estimate, never the suitability composite — a crowded district is not thereby an unsuitable one.

I Platform capabilities

- **Plan mode** fixes the study reading (mainland versus full administrative extent) and the composite rule before the run.
- **Analysis tools** cover fuzzy membership functions, raster algebra, focal-statistics repair, terrain and wetness indices, grids and zonal statistics in one chain.
- **Charts** mount the "which constraint decides the class" donut on the map canvas.
- **Reports** carry the threshold table, the coverage and repair table, the class shares and the district pressure list.

I Run it

1 Open the scenario

Choose **Carrying Capacity Assessment** in the scenario catalog. Six layers import and the camera flies to Jakarta (106.85, -6.21).

2 Send the suggested question

"Run a resource-and-environment carrying-capacity study for Jakarta: assess how water, climate and flood constraints limit growth, flag overloaded areas, then recommend measures by dominant constraint; deliver layers, charts and a report."

3 Confirm the method

The plan asks whether to read the main-city extent or the full administrative area including the offshore islands, whether to use the strict **minimum** composite (the default) or a compensating fuzzy gamma when a class degenerates, and which numbers the report should quote.

4

Approve and watch the run

After approval, each step runs as a deterministic analysis. The plan graph shows the four membership rasters feeding the limiting-factor composite and the pressure quadrant.

5

Read the failure modes in the result

Check that every class is non-empty and that the delivered raster covers at least 99% of the study area — that is the method's acceptance floor, not a promise about your numbers. If a class collapses, the documented fix is to calibrate the membership scale or run the gamma variant — not to renumber the classes.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped Jakarta data, the deliverables were:

- **99.1%** coverage of the study area after repairing both input gaps (91.9% before the repair), with the NoData patches reduced from thousands of speckles to a small remainder;
- four non-empty classes of roughly **11 / 23 / 25 / 41%** of the study area, where class 4 is the most constrained;
- a dominant-constraint reading of **wetness 57%, rainfall 33%**, with flood depth and slope accounting for the remainder;
- overload cells bringing together low capacity and high population, plus the per-district pressure table;
- one report and a mounted constraint-composition chart.

The population column should total the study-area population within about $\pm 10\%$ of the published census figure; a much larger sum points to a unit or resolution bug rather than a finding.

Read the results with care

- ▲ **Rainfall is a constraint, not a resource.** The scenario treats wetter as more limiting, following the waterlogging reading used in territorial appraisal.

▲ **Population is never a suitability factor.** It is exposure and pressure only.

▲ **The 300 m drainage corridor is an analysis default.** It degrades cells inside the corridor and must be labelled as an assumption.

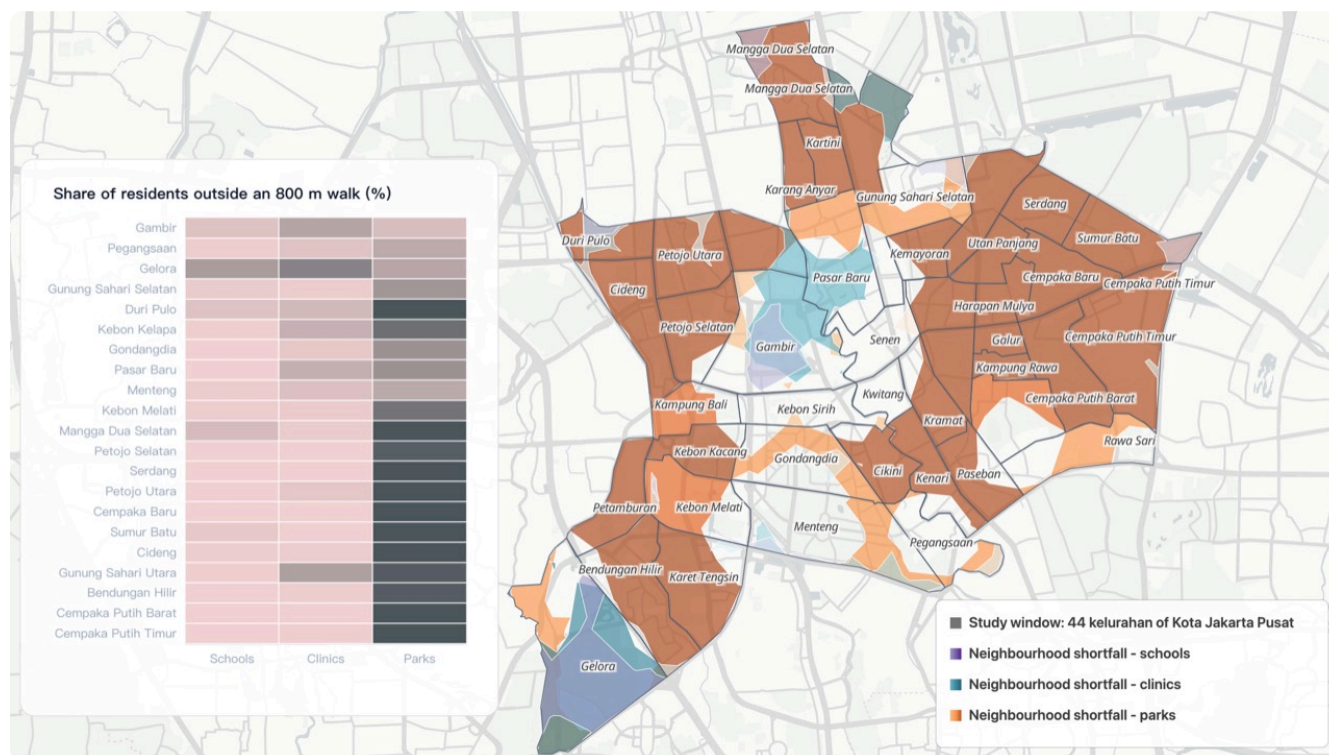
▲ **Two gaps are repaired and disclosed.** Never impute zero — zero falsely asserts "no flood". The report names what was filled and how much area it affected.

▲ **Land subsidence is out of scope.** The catalog ships no subsidence dataset, so the report says "subsidence not modelled" instead of substituting another raster.

▲ **Rainfall resolution is coarse.** At ~5.35 km it is a background constraint; quote it as such.

15-minute Life Circle Coverage Jakarta

Audit who can reach a school, a clinic and a park within an 800 m walk in central Jakarta, rank the neighbourhoods with the biggest shortfall, and score accessibility.



The Jakarta life-circle run as delivered: three neighbourhood shortfall ramps, ADM4 boundaries and the pinned three-class comparison heatmap

AT A GLANCE	
STUDY AREA Jakarta	ENTRY LAYERS 7 layers
KEY METHODS Walking service areas · shortfall population · 2SFCA	DELIVERABLES Three shortfall layers, a neighbourhood table, boards and a heatmap, a report

15-minute Life Circle Coverage measures the daily-life side of planning: **who can reach a school, a clinic and a park on foot**, and which neighbourhoods carry the largest shortfall. It is the MapX version of a 15-minute life-circle audit, and it deliberately reports two rulers — coverage and accessibility — instead of a single score.

What the scenario ships

Seven prepared layers cover Jakarta; the scenario's default study window is Kota Jakarta Pusat (central Jakarta).

Layer	Data	Role in the study
Education facilities	OSM education POIs (7,470 city-wide)	Schools that supply the education circle
Health facilities	Healthsites.io (2,447 city-wide)	Clinics and hospitals that supply the health circle
Road network	Overture transportation	The walking network used for service areas
Administrative boundaries	46 kecamatan (ADM3)	District framing and attribution
Green space and parks	OSM green polygons (3,942)	Source of the destination-park set
Neighbourhood boundaries	kelurahan (ADM4); 44 in the study area, 1.09 km ² average	The evaluation unit that is ranked and named
Population	WorldPop, 100 m	Residents at risk of being outside a circle

All layers come from the **built-in datasets** catalog with styles and attribution attached.

How the study works

Stage	Question	What MapX runs	Output
1. Coverage audit	How much of each neighbourhood is outside the walk?	Select features, Overlay clip/dissolve/erase, Network service area at 800 m, Zonal statistics, Field calculator	Shortfall layer per class with <code>gap_sum</code> and a share-of-residents column
2. Service mix and balance	Which neighbourhoods lack one, two or all three classes?	Zonal summary chaining, Overlay dissolve/erase	Neighbourhood summary table with population, coverage count 0–3 and three shortfall columns
3. Accessibility diagnosis	Where is access hardest, even inside the circle?	2SFCA accessibility, Top N ranking, Label anchors, Charts	Top-10 shortfall boards, per-class accessibility scores and a pinned heatmap

One radius makes the classes comparable. All three classes use an 800 m network walk, the upper end of the 15-minute life-circle standard (GB 50180-2018, 800–1000 m). A class-standard sensitivity — schools 800 m, clinics 1000 m, parks 500 m — is available as a documented variant.

The park definition is a decision, not a detail. The shipped green layer contains road verges, golf fairways and tiny slivers. The scenario keeps `leisure = park` polygons of at least 0.5 ha and uses one centroid per park (with the default definition and the shipped data, 33 destination parks in the study area). Relaxing that definition changes the answer dramatically, so the definition is printed in the report.

Coverage and accessibility answer different questions. The shortfall layer says whether residents can walk there; the 2SFCA score says how much supply there is per resident within reach. When they disagree, that disagreement is the finding.

Platform capabilities

- **Plan mode** asks the four decisions that matter — scope, radius, capacity assumption and deliverables — then freezes the rest.
- **Network analysis** builds the walking service areas; overlay and zonal tools turn them into neighbourhood-level numbers.
- **Charts** include per-class rankings, a grouped bar with a log axis and a heatmap, which can be pinned to the map while you pan.
- **Layer styling** keeps three semi-transparent ramps readable at once, with the ADM4 boundary and labels on top.
- **Reports** carry the radius, the park definition, the ranking tables and the caveats.

Run it

1 Open the scenario

Open **15-minute Life Circle Coverage** from the catalog. Seven layers import and the camera flies to Jakarta (106.85, -6.21).

2 Send the suggested question

"Run a 15-minute life-circle diagnosis for central Jakarta (Jakpus): check the walking service areas of schools, clinics and parks, find coverage gaps and imbalanced districts, and name the areas to prioritise; deliver coverage layers, charts and a report."

3

Answer the four clarifications

The plan asks for the study unit and facility classes (kelurahan plus all three classes by default), the walking radius (uniform 800 m), the 2SFCA supply assumption (default capacity 1, a pure supply/demand ratio) and the report language. Nothing else needs re-asking.

4

Approve the plan

After approval the run executes as a plan graph: filter the study area, clip the three facility classes, build the 800 m service areas, erase them from the neighbourhoods and aggregate the results.

5

Check the supply side before reading conclusions

The run reports how many facilities entered each class. In one example run on the shipped data, the study window filtered about 601 education points, 234 health points and 33 parks; compare those figures with the counts that entered your run — a class showing one point where hundreds entered means a broken chain, not a finding.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped central-Jakarta data, the deliverables were:

- three neighbourhood shortfall ramps (schools in purple, clinics in teal, parks in orange) sharing one 0/10/25/50/75/100% break scale, with the ADM4 boundary above them — exactly four visible deliverable layers;
- a 44-row neighbourhood summary table carrying population, coverage count 0–3 and the three shortfall columns;
- per-class top-10 boards and 2SFCA accessibility layers, hidden by default;
- five charts: three per-class rankings, one grouped bar over the worst neighbourhoods, and a pinned heatmap comparing all three classes per kelurahan;
- a report naming the radius, the park definition and the priority neighbourhoods.

That example also shows why both the definition and the network matter: the same 33 parks cover about **32%** of residents through the road network, while a straight-line ring from the park boundary suggests about **56%**. Widening the definition to every `leisure = park` polygon with no size filter takes the straight-line reading to about **92%**, against **49%** for the same 33 parks. Straight-line

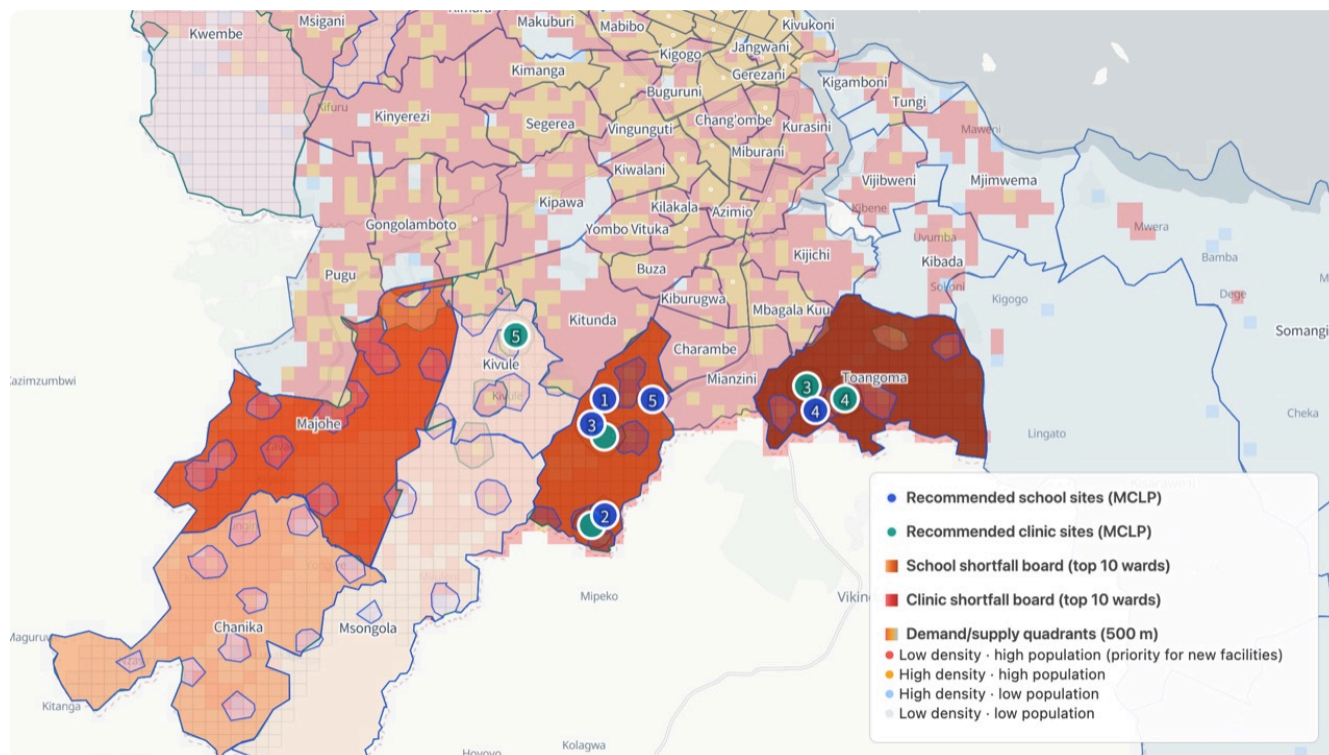
proximity overstates walking access, and the exact shares depend on the park definition and the network data version.

Read the results with care

- ▲ **Coverage is a network distance.** When the mapped street network is incomplete, the gap is overestimated; the report says so.
- ▲ **A park is represented by its centroid.** The 800 m distance is measured to the park's centre, an accepted approximation when gate data is unavailable.
- ▲ **Blank means inside the circle.** In the shortfall layers, unshaded neighbourhoods are covered — they are not missing data.
- ▲ **cov_count counts intersection, not full coverage.** A neighbourhood that a circle only clips counts as covered; quote the covered area when a borderline case matters.
- ▲ **Capacity is not modelled.** The 2SFCA score uses a default capacity of 1, so it ranks within a class only; cross-class comparison uses the shortfall population.
- ▲ **The answer is neighbourhood-level.** "Where inside one kelurahan should we build?" is the **Public Facility Siting** scenario.

Public Facility Siting Dar es Salaam

Diagnose school and clinic coverage, measure the unserved demand, then compare candidate sites and recommend where to build with a maximum-coverage model.



The Dar es Salaam run as delivered: ward shortfall boards, demand/supply quadrants and ranked MCLP sites for schools and clinics

AT A GLANCE	
STUDY AREA Dar es Salaam	ENTRY LAYERS 7 layers
KEY METHODS Coverage diagnosis · candidate screening · MCLP vs k-median	DELIVERABLES Shortfall boards, candidate cells, recommended sites, a marginal curve and a report

Public Facility Siting answers the last mile of public-service planning: **given what already exists, where should the next schools and clinics go?** The scenario runs the full arc — coverage diagnosis, demand gap, candidate screening and siting comparison — and reports how many currently unserved residents each recommended site would bring within reach.

What the scenario ships

Seven layers cover the Dar es Salaam region. The watercourse layer starts hidden; it exists so the 30 m river setback can be applied to candidate sites.

Layer	Data	Role in the study
Education facilities	OSM education POIs (2,223)	Raw supply for the school analysis
Health facilities	Healthsites.io (1,463)	Raw supply for the clinic analysis
Road network	OSM roads (102,464 lines)	Walking/driving network for service areas and site snapping
Watercourses	OSM waterways (11,160 lines, hidden)	30 m riparian setback applied to candidates
Ward boundaries	OCHA COD-AB ADM3 (90 wards, 1,667 km ²)	Shortfall ranking, candidate scope and balance checks
Building footprints	Overture (481,681 polygons)	Built-up intensity screen for candidate cells
Population	WorldPop, 100 m	Demand and coverage weights

How the study works

Stage	Question	What MapX runs	Output
1. Coverage diagnosis	Who is outside the standard today?	Select features (class screen), Overlay clip, Service area (800 m schools / 1000 m clinics), Overlay erase, Zonal statistics, Top N	Ward shortfall layer and a top-10 board per class
2. Demand gap	Where is demand high and supply low?	Grid at 500 m, Zonal statistics, Quadrant (median split), 1 km demand grid	Pressure quadrants (diagnostic) and the unserved-population demand layer
3. Siting comparison	Which new sites serve the most people?	Grid inside the worst wards, Overlay erase (30 m corridor), Zonal statistics/summary, Centroid points, Location-covering (MCLP), Location-allocation (k-median)	Recommended sites, the efficiency comparison, marginal coverage curve and residual gaps

The supply set is narrowed before anything is computed. Of 2,223 education points, 1,312 are schools — kindergartens, colleges and universities are not school supply for a walking-radius question. Of 1,463 health

points, only 363 are clinics, hospitals or doctors; **1,020 entries (69.7%) are pharmacies** and are excluded. Without that screen, a "clinic" analysis is really a pharmacy map.

Candidates are a shortlist, not the whole city. The scenario ranks the 90 wards by unserved population, keeps the ten worst, lays a 500 m grid inside them, removes the 30 m river corridor, drops cells that are already built out (building footprint share above 30%) and then caps the survivors by residential population. The ladder for large windows is documented; the physical filters are never relaxed to fit a budget.

Two models, one recommendation. The primary model is maximum coverage (MCLP): place k sites so the most unserved people fall within the standard radius. A k -median run with the same candidates and demand is reported alongside as the efficiency comparison — it minimises travel, but it is not the recommendation.

Platform capabilities

- **Network analysis** runs `service_area`, `location_covering` and `location_allocation`, with the platform's scale guard reporting the effective candidate limit before submission.
- **Plan mode** records the supply-set definitions, the radius per class, the demand basis and the `k` as plan assumptions.
- **Charts** plot the marginal coverage curve so you can see whether the fourth or fifth site still earns its place.
- **Reports** carry the before/after coverage pair and every disclosure listed below.

Run it

1

Open the scenario

Open **Public Facility Siting** from the catalog. Seven layers import and the camera flies to Dar es Salaam (39.21, -6.79).

2

Send the suggested question

"Run a public-facility siting study for Dar es Salaam: diagnose the coverage and demand gaps of existing schools and clinics, then compare candidate sites for new facilities and recommend where to build; deliver layers, charts and a report."

3

Confirm the decisions

The plan asks which facility class you are adding, the study area (the whole region or named wards), the primary model (MCLP by default, with k -median as the comparison), the number of sites (`k`, default 5), the service radius and the report language.

4

Approve and read the scale note

The submit response states the effective candidate limit and which axis binds it — candidates, demand points or runtime. If the shortlist exceeds it, the documented ladder coarsens the grid or shrinks the ward set; it never drops the feasibility filters.

5

Compare the models and deliver

Read the marginal coverage curve, name the worst wards that received no site, then export the report or share the map with the recommended sites.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped Dar es Salaam data, the deliverables were:

- **90 wards** as the screening and reporting unit, with the status-quo service areas at 800 m for schools and 1000 m for clinics;
- supply counts of **1,312 schools (1,305 inside the study area)** and **363 clinics (362 inside)**, with the excluded pharmacy share stated in the report;
- **five recommended school sites and five recommended clinic sites** ($k = 5$) placed only inside the worst ten wards, with their service areas and the residual gaps after siting;
- the k-median comparison sites, a ward balance table naming wards with no site, and a marginal coverage curve;
- a before/after population pair: unserved residents at the start versus residents brought within reach by the new sites.

Read the results with care

- ▲ **Capacity is not modelled.** The clinic layer has no bed or staff counts and the school layer has a capacity value on only 15 of 2,223 features, so the run is plain maximum coverage; the report writes "capacity not modelled".

▲ **Coverage is binary.** "Within 800 m of a school" is not "able to enrol". Run the 2SFCA crowding branch if you need a supply-per-resident reading.

▲ **Construction intensity is footprint share only.** Only 34 of 481,681 building footprints carry a height value, so never infer floors or population capacity from the building layer.

▲ **A recommended site is a candidate-cell optimum.** Land ownership, zoning and site access still need a field check.

▲ **Fairness is not optimised.** MCLP maximises the total; the ward balance table exists so wards that received nothing are named rather than hidden.

▲ **The ward layer is the 2018 administrative vintage.** Dar es Salaam is still the three-council structure in this release; do not describe it as the current five-municipality structure.

Urban Growth Monitor **Hanoi**

Reconstruct Hanoi's built-up expansion phase by phase, measure how much is infill versus outward growth, and connect growth to population and night lights.



The Hanoi run as delivered: infill, edge and outlying expansion forms over the built-up raster, with population and night-light quadrants

AT A GLANCE	
STUDY AREA Hanoi	ENTRY LAYERS 7 layers
KEY METHODS Phased expansion metrics · landscape expansion index · centre and ellipse	DELIVERABLES Expansion-form layers, phase tables, quadrant grid, charts and a report

Urban Growth Monitor asks three questions a growth-management plan needs to answer: **how fast is the city expanding, where is it going, and is the growth compact or sprawling?** It reconstructs the expansion phase by phase, separates infill from edge and outlying growth, and overlays population and night lights to find where growth and activity disagree.

What the scenario ships

Seven layers cover the Hanoi window. Only the built-up area is visible by default; the rest are context and analysis inputs.

Layer	Data	Role in the study
Road network	OSM roads (hidden)	Test whether new built-up land follows the main axes
Watercourses	OSM waterways, 2,565 lines (hidden)	River corridors as a growth constraint
Administrative boundaries	50 ADM3 units (hidden)	Attribution and cross tables
Built-up area	GHSL 2020 density, max 7,967 m ² /cell (visible)	Growth-intensity backdrop
Built-up history	WSF Evolution 1985–2015 build year, 733k valid cells (hidden)	The phase and form source
Population	WorldPop 2020 (hidden)	Demand and activity axis
Night lights	VIIRS, December 2017 median, 611 m (hidden)	Economic-activity proxy

How the study works

Stage	Question	What MapX runs	Output
1. Reconstruct the expansion	How much built-up land was added in each period?	Expansion metrics (<code>periodBreaks</code> 1990 / 2000 / 2010 / 2015), Raster class area, Land change matrix	Phase table with new area, annual growth, intensity and compactness; built-year cross table
2. Measure the form	Is new growth infill, edge expansion or outlying?	Landscape expansion index on two built-up masks, Raster to polygons, Mean centre, Standard deviational ellipse, Directional distribution, Buffer and zonal summary	Infill / edge / outlying layer, centre shift and ellipse, share of new built-up near roads and waterways
3. Connect to activity	Where do population and night lights diverge?	Grid at 1 km, Zonal statistics, Quadrant (median split)	Population × night-light quadrants and the growth-management targets

Form is measured with the landscape expansion index (LEI), not density. Density cannot tell a dense low-rise extension from a dense infill block; the LEI separates them by how much of a new patch touches the earlier built-up area. The scenario uses the ≤ 1990 and ≤ 2015 masks (one-cell neighbourhood, 50% threshold, four-cell minimum patch) and reports the shares with those parameters attached.

The centre and the ellipse describe direction. The gravity centre and its standard deviational ellipse are computed per phase, so the report can say how far and in which direction the city's mass moved, and whether it stretched along one axis.

Population and night lights are read together. Cells with high population but low night light are the classic mismatch — a growth-management priority rather than a success story.

I Platform capabilities

- **Analysis tools** include expansion metrics and the landscape expansion index, which are built for this kind of multi-epoch reading.
- **3D view** is optional: the built-up density raster can be extruded for a quick sense of the vertical story.
- **Charts** combine a dual-axis bar-and-line chart, a donut for the form shares and a bar for the quadrant distribution.
- **Reports** carry the phase table, the form and centre numbers, and the year-mismatch caveat.

I Run it

1 Open the scenario

Open **Urban Growth Monitor** from the catalog. Seven layers import and the camera flies to Hanoi (105.83, 21.03) on a dark basemap.

2 Send the suggested question

"Run an urban-growth monitoring study for Hanoi: reconstruct the staged expansion of built-up land, analyse its speed, direction and intensity, flag the main growth-management issues with boundary guidance; deliver layers, charts and a report."

3 Confirm the reading

The plan asks for the study area and admin level, whether the expansion reading covers the whole city or a ring around the centre, the two mask years (defaults ≤ 1990 and ≤ 2015), and the numbers to quote in the report.

4

Approve and read the phase table

After approval, the run publishes the phase metrics first, then the LEI form layer and the population/night-light quadrants. Quote the infill share with its neighbourhood and mask parameters.

5

Use the findings

The quadrant layer names the cells to steer; the form layer shows whether policy should focus on infill capacity or on servicing outward growth.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped Hanoi data, the deliverables were:

- four phases of **193.0 / 147.8 / 252.4 / 66.8 km²** of new built-up land, summing to the 660.0 km² in the raster;
- **+467.0 km²** of new built-up land between 1991 and 2015, against a 469.1 km² LEI total — a 0.4% difference between the two area conventions that the report states;
- expansion forms of **2.1% infill, 27.4% edge expansion and 70.5% outlying** — in that example, most new built-up land was outlying;
- a gravity centre that moved from (105.8186, 21.0301) to (105.8114, 21.0133) — about **2.0 km south-southwest** — while the 1 σ major axis grew from 9.58 km to 11.06 km;
- population/night-light quadrants of **1,311 / 657 / 657 / 1,310** cells out of 3,935 (medians: 932.6 residents, 1.96 night-light units);
- 2,181 new built-up patches with a mean size of 239 cells (21.5 ha), plus three charts and a report.

Read the results with care

- ▲ **The sources have different reference years.** Build year is ≤ 2015 , built-up density and population are 2020, night lights are December 2017. Any cross-source regression is a trend indication only.

▲ **Only one land-cover vintage is shipped.** Do not claim which land-use category was converted into built-up land.

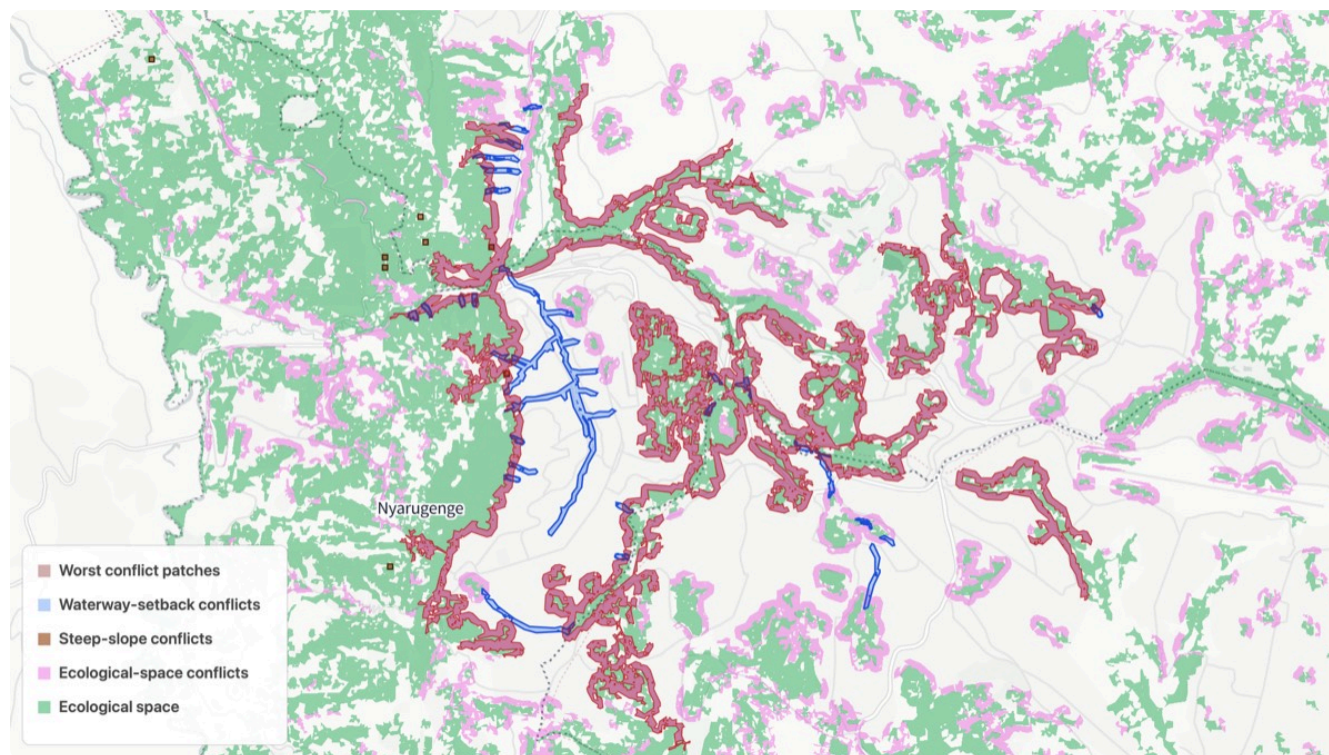
▲ **Compactness is a per-pixel shape metric.** A raster mask's staircase edge dominates its perimeter; compare it only within the same raster and resolution, never across cities.

▲ **LEI depends on its parameters.** Quote the neighbourhood size, threshold and mask years with the infill/edge/outlying shares. Cells on the raster edge are counted as non-built, so the boundary band skews towards edge and outlying.

▲ **"New patches" are clusters of a class mask.** For patch structure, quote the whole-new-built patch count from your run (2,181 patches, mean 21.5 ha, in the example run).

Land-use Conflict Screening **Kigali**

Find where Kigali's development presses on ecological space, steep slopes and riverbanks, grade the conflicts, and turn them into remediation priorities.



The Kigali run as delivered: ecological space with the three conflict families and the worst-patch board

AT A GLANCE	
STUDY AREA Kigali	ENTRY LAYERS 8 layers
KEY METHODS Conflict screening · grading and prioritisation · remediation guidance	DELIVERABLES Four conflict layers, a worst-patch board, an exposure grid and a report

Land-use Conflict Screening checks whether a hilly city's built-up land collides with the constraints it should respect: **ecological space, steep slopes, riverbanks and protected areas**. It reports each conflict family on its own and then a de-duplicated union, so one patch cannot be counted twice, and it ends with remediation priorities instead of a single "risk score".

What the scenario ships

Eight layers cover the Kigali window. Only land cover is visible by default; the other layers are analysis inputs and cross-checks.

Layer	Data	Role in the study
Watercourses	OSM waterways, 416 lines (hidden)	30 m setback corridor
Administrative boundaries	8 ADM3 units (hidden)	Study area, attribution and labels
Protected area constraints	Overture, 14 features in Rwanda; 1 inside the city (hidden)	Background and cross-check for ecological space
Building footprints	Overture, 644,665 polygons (hidden)	Context layer; not used as the development surface (see below)
Land cover	ESA WorldCover 2021, 10 m, 11 classes (visible)	Ecological classes and the built-up surface
Terrain	SRTM 30 m, clipped to the city (hidden)	Slope classes
Surface water	JRC Global Surface Water, 0–99% frequency (hidden)	Water presence cross-check
Population	GHSL, 100 m (hidden)	Exposed population per assessment cell

How the study works

Stage	Question	What MapX runs	Output
1. Identify the conflicts	Where does built-up land meet each constraint?	Raster calculator, Raster to polygons, Overlay clip, Euclidean distance (100 m / 30 m), Buffer, Slope, Reclassify	Ecological-space mask, steep-slope classes and the four conflict families
2. Grade and prioritise	Which conflicts matter most, and where do they cluster?	Append features, Zonal summary on a 500 m grid, Top N patch ranking, Label anchors	Conflict area per cell, exposed population, worst-patch board
3. Remediate	What should be done per type and severity?	Select features, Zonal statistics, Charts, Report	Remediation table, charts and a compliance report

Ecological space is a composite, because the protected layer is nearly empty here. Inside the city, the protected-area layer contains a single unnamed feature of about 0.09 km². The working definition is therefore the trees, shrub, water and wetland land-cover classes (WorldCover 10 / 20 / 80 / 90), with protected areas kept as a background cross-check and the 30 m waterway corridor as a separate constraint.

Development is read from the built-up land-cover class, not from building footprints. The shipped footprint layer has 644,665 polygons; on this data, clipping, intersecting and aggregating it all exceed the platform's step budget. The runnable surface is the built-up class of the same 10 m land-cover raster, and the report states the substitution and the reason.

Conflict families are reported per type plus a union. A patch can sit on a steep slope *and* inside the waterway setback. The scenario reports each family's area, then the de-duplicated union — never a naive sum presented as "total conflicts".

I Platform capabilities

- **Analysis tools** combine raster algebra, distance thresholds, buffers, overlays and zonal aggregation in one chain.
- **Plan mode** records the ecological-space definition, the development surface and the union rule as explicit assumptions.
- **Layer styling** gives each constraint family its own hue so four translucent layers stay readable together.
- **Reports** carry the definitions, the per-type and union areas, and the remediation table.

I Run it

1

Open the scenario

Open **Land-use Conflict Screening** from the catalog. Eight layers import and the camera flies to Kigali (30.06, -1.94).

2

Send the suggested question

"Screen land-use conflicts for Kigali: find development encroaching on ecological space, steep slopes or riverbanks, grade conflicts by severity, locate hotspots and recommend fixes; deliver layers, charts and a report."

3

Confirm the definitions

The plan asks which constraint types count as conflicts (all four by default), whether the development surface may be the built-up land-cover class (the runnable default), and how the families should be combined — per type plus a union.

4

Approve and read the per-type numbers

After approval, the run vectorises the ecological patches, builds each conflict mask, and aggregates them onto the 500 m grid with population.

5

Treat an empty family as a finding

A constraint family with zero features is a legitimate result. Report "0 patches" and what it means; do not tune parameters to force a non-empty layer.

Example output

NOTE

The values below come from one example run with the default settings on the shipped data. They show the shape of the deliverables, not what your run will return — your window, parameters and data updates will change them.

In one example run on the shipped Kigali data, the deliverables were:

- an ecological-space layer of **1,387 patches / 270.1 km²** (patches of at least 1 ha), with the city mask applied to the vectorised patches rather than to the raster;
- a terrain layer clipped from the country-wide DEM to **435 × 436** cells before slope analysis;
- the four conflict families as separate layers — development on $\geq 25^\circ$ slopes (8 cells / 0.046 km² in that example run), inside the 30 m waterway setback, within 100 m of ecological space, and inside protected areas (no built-up land inside the in-city protected feature);
- a 500 m grid with conflict area and exposed population, plus a worst-patch board;
- a report with the per-type and union areas and the remediation guidance.

Read the results with care

- ▲ **The in-city protected layer is nearly empty.** It cannot carry the ecological definition alone; the composite and the cross-check are both reported.

▲ **No building height, no volume.** The footprint layer carries a height value on almost none of its features, and on this data it falls outside the step budget — the report says development is read from land cover.

▲ **No slope-stability data.** Do not output a landslide risk class; the scenario gives "steep slope + setback + foundation treatment" guidance instead.

▲ **Report areas per type plus a union.** One developed patch can hit several constraints; a sum would double-count it.

▲ **The 100 m encroachment band is an analysis default.** It expresses "development pressing on ecological space" because one reference year cannot contain a cell that is both built-up and ecological.

▲ **Surface-water frequency is classed before use.** Raw 0–99% frequency is not a binary constraint.

APPENDIX A

Appendix A: Scenarios at a glance

Scenario	Study area	Layers	Key methods	Deliverables
1 Development Suitability Baseline	Nairobi	8	Hard-constraint screening, weighted overlay, capacity estimate	Suitability classes, development control tiers, charts and a report
2 Carrying Capacity Assessment	Jakarta	6	Fuzzy membership, limiting-factor composite, pressure quadrants	Capacity classes, overload cells, charts and a report
3 15-minute Life Circle Coverage	Jakarta	7	Walking service areas, shortfall population, 2SFCA	Three shortfall layers, a neighbourhood table, boards and a heatmap
4 Public Facility Siting	Dar es Salaam	7	Coverage diagnosis, candidate screening, MCLP vs k-median	Recommended sites, a marginal coverage curve, a ward balance table
5 Urban Growth Monitor	Hanoi	7	Phased expansion metrics, landscape expansion index, centre and ellipse	Expansion-form layers, phase tables, a quadrant grid
6 Land-use Conflict Screening	Kigali	8	Conflict screening, grading, remediation guidance	Four conflict layers, a worst-patch board, an exposure grid

Appendix B: Get started

- Open the **Scenarios** catalog from the MapX top menu and pick the scenario you need.
- Choose **Open scenario**: the whole layer bundle imports into the current project with default styles, no upload or configuration required.
- Start the analysis with the scenario's **suggested question**, or edit it first.
- In **plan mode**, answer the clarifications, review the plan document and approve it.
- When the run finishes, export a report, work with the charts or share the map; keep asking follow-up questions to adjust the scope and thresholds.

Appendix C: Data and licences

Scenarios are assembled from the built-in data catalog, and every layer carries its source, licence and attribution. Land cover, population, built-up area, protected areas and road networks each have their own vintage and resolution; state the year and scale whenever you combine them. Scenario output is analysis, not a statutory planning document.



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